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B. Tech. Degree V Semester Examination November 2014**ME 1503 MACHINE TOOLS AND MACHINING SCIENCE**
(2012 Scheme)

Time: 3 Hours

Maximum Marks: 100

PART A
(Answer *ALL* questions)

(8 x 5 = 40)

- I. (a) What are the functions of feed rod and lead screw?
(b) How do you get the reciprocating motion in a slotting machine?
(c) Discuss the importance of the rake angle in milling.
(d) Describe the constant factors and variable factors in the selection of grinding wheels.
(e) What are the important assumptions made in orthogonal machining and why they are significant?
(f) Sketch a peripheral milling cutter and mark important tool angles.
(g) What are the requirements of a good tool material?
(h) Why are chip breakers provided on the rake face?

PART B

(4 x 15 = 60)

- II. Explain the construction of a radial drilling machine with emphasis on how the requisite motions are obtained.

OR

- III. What are the features of a boring machine which make it an accurate production machine?

- IV. (a) Describe a grinding wheel structure, with the help of a neat sketch.
(b) Mention the various types of bonds used in the making of grinding wheels.

OR

- V. (a) Explain the advantages of employing jigs and fixtures in mass production.
(b) Describe the principles of jigs and fixtures design.

- VI. (a) Develop a relationship between shear strain, shear angle and rake angle.
(b) In an orthogonal machine operation, the rake angle of the tool was 15° , shear angle 35° . Calculate shear strain.

OR

- VII. (a) Why does the temperature in cutting depend on the cutting speed, feed, and depth of cut? Explain in terms of the relevant process variables.
(b) Define shear plane angle, rake angle and clearance angle in orthogonal machining.

- VIII. (a) What factors contribute to costs in machining operations?
(b) Would you expect that a wear equation for crater wear would be substantially different? Explain.

OR

- IX. (a) Explain the relationship between shear angle, angle of friction and cutting rake angle developed by Merchant.
(b) Discuss the forms of tool life equations generally used with their applicability.
